Trainings

Preparatory Steps

with Mechanically Actuated Injector

⚠ WARNING ⚠

Do not remove the pressure cap from a hot engine. Wait until the coolant temperature is below 50°C [120°F] before removing the pressure cap. Heated coolant spray or steam can cause personal injury.

⚠ WARNING ⚠

Coolant is toxic. If not reused, dispose of in accordance with local environmental regulations.

⚠ WARNING ⚠

This component or assembly weighs greater than 23 kg. [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift component or assembly.

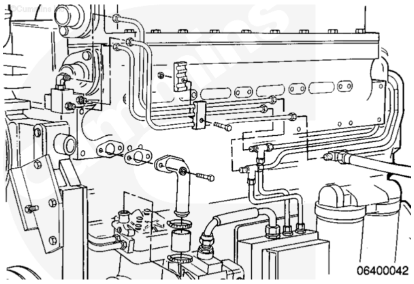
⚠ CAUTION ⚠

Do not spill or drain fuel into the bilge area when disconnecting or removing fuel lines, replacing filters, and priming the fuel system. Do not drop or throw filter elements into the bilge area. The fuel and fuel filters must be disposed of in accordance with local environmental regulations.

- Disconnect the battery cables. Refer to Procedure 013-009 in Section (/qs3/pubsys2/xml/en/procedures/20/20-013-009.html)
- Remove the engine wiring harness. Refer to Procedure 019-043 in Section (/qs3/pubsys2/xml/en/procedures/13/13-019-043.html) in the Troubleshooting and Repair Manual, Electronic

Control System, QSK19 CM850, Modular Common Rail System Series Engine, Bulletin 4021493.
- Remove the fuel supply tubes. Refer to Procedure 006-024 in Section (/qs3/pubsys2/xml/en/procedures/20/20-006-024-tr.html)
- Drain the cooling system. Refer to Procedure 008-018 in Section (/qs3/pubsys2/xml/en/procedures/20/20-008-018-tr.html)
- Remove the aftercooler assembly. Refer to Procedure 010-002 in Section (/qs3/pubsys2/xml/en/procedures/20/20-010-002-tr.html)

On engines equipped with an air compressor, remove the air inlet and outlet connections, from the air compressor. Refer to Procedure 012-014 in Section (/qs3/pubsys2/xml/en/procedures/20/20-012-014-tr.html)



Remove

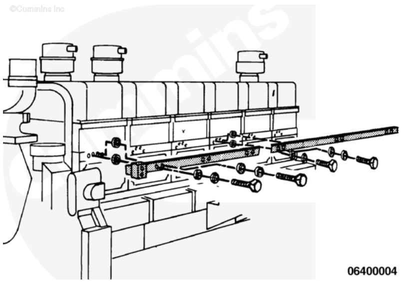
with Mechanically Actuated Injector

Remove the 12 capscrews.

Remove the **front** fuel manifold.

Discard the o-rings.

Repeat this process for the removal of the **rear** fuel manifold.



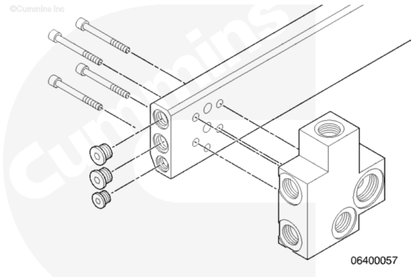
Disassemble

with Mechanically Actuated Injector

Remove the four fuel connection block capscrews.

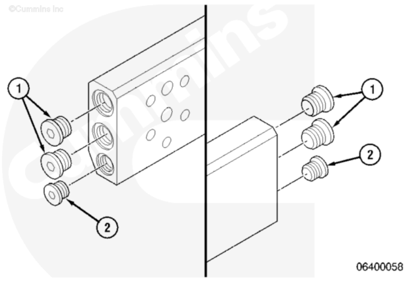
Remove the fuel connection block.

Discard the o-rings.



Remove the two 3/8 inch plugs (2) and four 1/2 inch plugs (1) at the end of the manifold.

Discard the o-ring if damaged or enlarged.



Clean and Inspect for Reuse

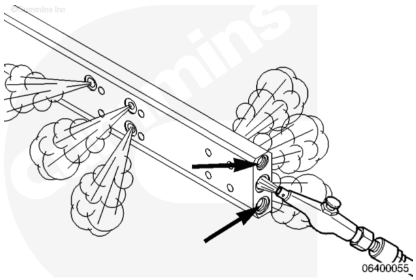
with Mechanically Actuated Injector

⚠ WARNING ⚠

When using solvents, acids or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.

⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.



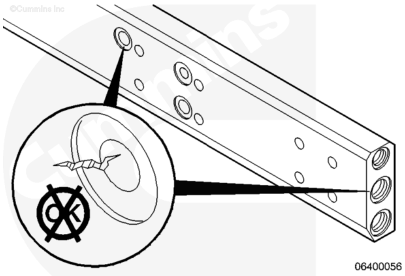
Use solvent to clean the manifold.

Use compressed air to clean all of the drillings.

Be sure all the drillings are clean and open.

Check the manifold for cracks in the fuel passages.

The manifold **must** be replaced if damaged.

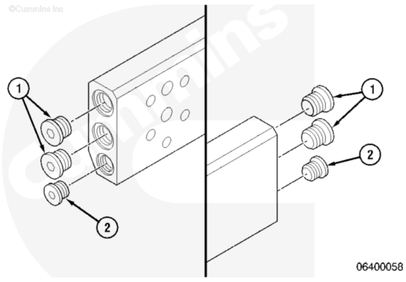


Assemble

with Mechanically Actuated Injector

Use clean engine oil to lubricate the o-rings.

Install the 3/8 in (9/64 in hex socket) (2) and 1/2 in plugs (3/16 in hex socket) (1).



Torque Value:
Plug (1):

1. 20 n•m [177 in-lb]

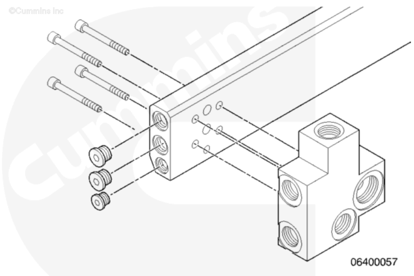
Torque Value:
Plug (2):

1. 7 n•m [62 in-lb]

Use Lubriplate® to secure the o-rings in position.

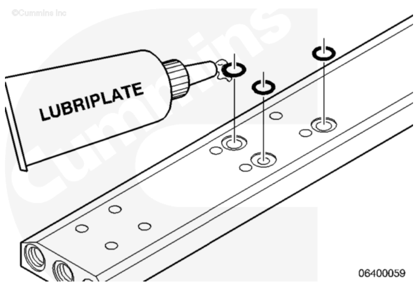
Install the fuel connection block and the four new o-rings.

Torque Value: 10 n•m [89 in-lb]



Install

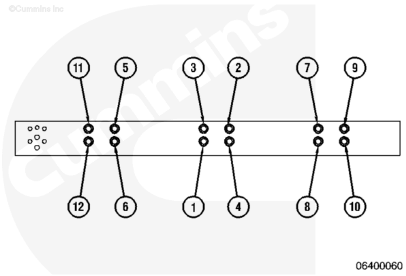
Use Lubriplate® Number 105, or equivalent, to position and secure the o-rings. Install the o-rings.



Install the **front** fuel manifold. Use the sequence illustrated to tighten the 12 capscrews.

Torque Value: 10 n•m [89 in-lb]

Repeat this process for the installation of the **rear** fuel manifold.



Finishing Steps

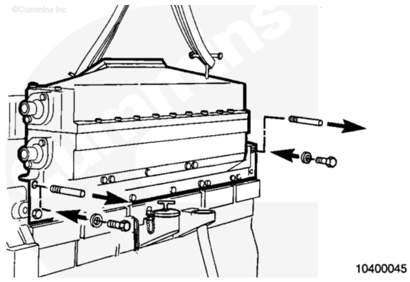
with Mechanically Actuated Injector

⚠ WARNING ⚠

Coolant is toxic. If not reused, dispose of in accordance with local environmental regulations.

⚠ WARNING ⚠

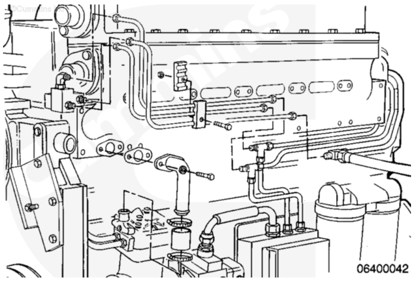
This component or assembly weighs greater than 23 kg. [50lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift component or assembly.



- Install the aftercooler assembly. Refer to Procedure 010-002 in Section (/qs3/pubsys2/xml/en/procedures/20/20-010-002-tr.html)
- Install the fuel supply tubes. Refer to Procedure 006-024 in Section (/qs3/pubsys2/xml/en/procedures/20/20-006-024-tr.html)
- Fill the cooling system. Refer to Procedure 008-018 in Section (/qs3/pubsys2/xml/en/procedures/20/20-008-018-tr.html)
- Install the engine wiring harness. Refer to Procedure 019-043 in Section (/qs3/pubsys2/xml/en/procedures/13/13-019-043.html) in the Troubleshooting and Repair Manual, Electronic Control System, QSK19 CM850, Modular Common Rail System Series Engine, Bulletin 4021493.
- Connect the battery cables. Refer to Procedure 013-009 in Section (/qs3/pubsys2/xml/en/procedures/20/20-013-009.html)

- Operate the engine and check for leaks.

On engines equipped with an air compressor, install the air inlet and outlet connections to the air compressor. Refer to Procedure 012-014 in Section (/qs3/pubsys2/xml/en/procedures/20/20-012-014-tr.html)



Last Modified: 20-Aug-2015